

## DEPI – Software Testing Track

### Part 01: Java Fundamentals

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## Session 04

### Assignment # 02

#### Java OOP Assignment – Soccer League System

Study the “Soccer League Games” table shown in the image. The table displays the results of matches between six teams, the total goals and points of each team, and a detailed view of one match that lists the players who scored and the minute of each goal.

#### Your Task

**Design a Java object-oriented solution using exactly the following five classes:**

1. Player
2. Team
3. Goal
4. Match
5. Schedule

#### For each class you must:

- List all necessary attributes (with their data types).
- List all necessary methods (constructors, getters, setters, and any other useful methods).
- Clearly indicate composition: state when one class is used as an attribute (or as an element of an ArrayList) inside another class.

#### Required Design Rules

- A Team must contain a list of its Players.
- A Match must contain two Teams (home and away) and a list of Goals.
- A Goal must store the match identifier, the Player who scored, and the minute the goal was scored.
- The Schedule class is responsible for holding all teams and all matches, updating team statistics (goals and points), ranking the teams, and displaying the results table and match details.

**Use ArrayList wherever a collection is needed.**

**Write a clear and complete class design that satisfies all the requirements above.**